

PES UNIVERSITY

Department of Computer Science & Engineering

LAB 1: REQUIREMENTS ENGINEERING & UML USE-CASE MODELLING

PROBLEM STATEMENT #12

Blood Bank Inventory & Emergency Donor Matcher

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Problem Statement

The Blood Bank Inventory & Emergency Donor Matcher is a real-time system intended to monitor blood-bank stock, track blood-component shelf lives, and support emergency donor matching during critical shortages. The system cross-references emergency blood requests against available inventory and can broadcast SMS alerts to compatible eligible donors within a 10 km radius.

Functional Requirements

The following five functional requirements define the core behavior of the proposed system.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall maintain real-time records of available blood units and their components, blood groups, quantities, and expiry dates.	High	Pass: Inventory information is updated whenever blood units are added, allocated, or removed. Fail: Inventory displays outdated or incorrect stock information.	Accurate real-time inventory is necessary to determine whether required blood is available during normal and emergency situations.
FR-002	Functional	The system shall monitor blood-unit shelf lives and identify blood units that are expired or approaching expiry.	High	Pass: The system identifies expired and near-expiry units according to configured shelf-life rules. Fail: An expired unit remains available for allocation.	Monitoring shelf life helps prevent the allocation of expired blood units.
FR-003	Functional	The system shall cross-reference emergency blood requests with available inventory to determine whether compatible blood units are available.	High	Pass: The system correctly identifies compatible available blood units for an emergency request. Fail: Incompatible or unavailable blood units are selected.	Emergency requests require rapid and accurate matching against current inventory.
FR-004	Functional	The system shall identify eligible compatible donors within a 10 km radius of the emergency blood request location and send them SMS notifications.	High	Pass: Eligible compatible donors within 10 km receive an alert. Fail: Ineligible or incompatible donors are notified.	Rapidly contacting nearby compatible donors can help address critical blood shortages.
FR-005	Functional	The system shall record and track emergency blood requests, donor notifications, and blood-unit allocations.	Medium	Pass: Each emergency request and related notification/allocation is recorded successfully. Fail: A request, notification, or allocation cannot be traced in the system.	Tracking activities provides a record of emergency operations and supports accountability.

Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The system shall maintain transactional consistency when allocating blood units and prevent the same blood bag from being allocated simultaneously to multiple requests.	High	Pass: Concurrent allocation tests confirm that a blood bag can be assigned to only one request. Fail: The same blood bag is allocated to multiple requests.	Preventing duplicate allocation is essential for maintaining accurate and safe blood inventory records.
NFR-002	Performance	The system shall send emergency donor notifications within 30 seconds of an emergency request being flagged.	High	Pass: Testing confirms that compatible eligible donors are notified within 30 seconds under the defined operating load. Fail: Notifications exceed the 30-second target.	Emergency blood shortages require rapid donor communication.

Actors

Actor	Role
Emergency Requester	Initiates an emergency blood request and receives the processing status.
Blood Bank Manager	Manages blood inventory, monitors expiry, handles emergency shortages, and oversees allocations.
Eligible Donor	Receives emergency donor notifications and can provide a response to a donation request.
SMS Notification Service	External service responsible for delivering SMS alerts to matching donors.

Primary Use Cases

- Manage Blood Inventory
- Monitor Blood Expiry
- Create Emergency Blood Request
- Check Blood Availability
- Match Compatible Blood
- Find Eligible Nearby Donors
- Send Emergency SMS Alert
- Record Donor Response
- Allocate Blood Unit
- Track Emergency Request

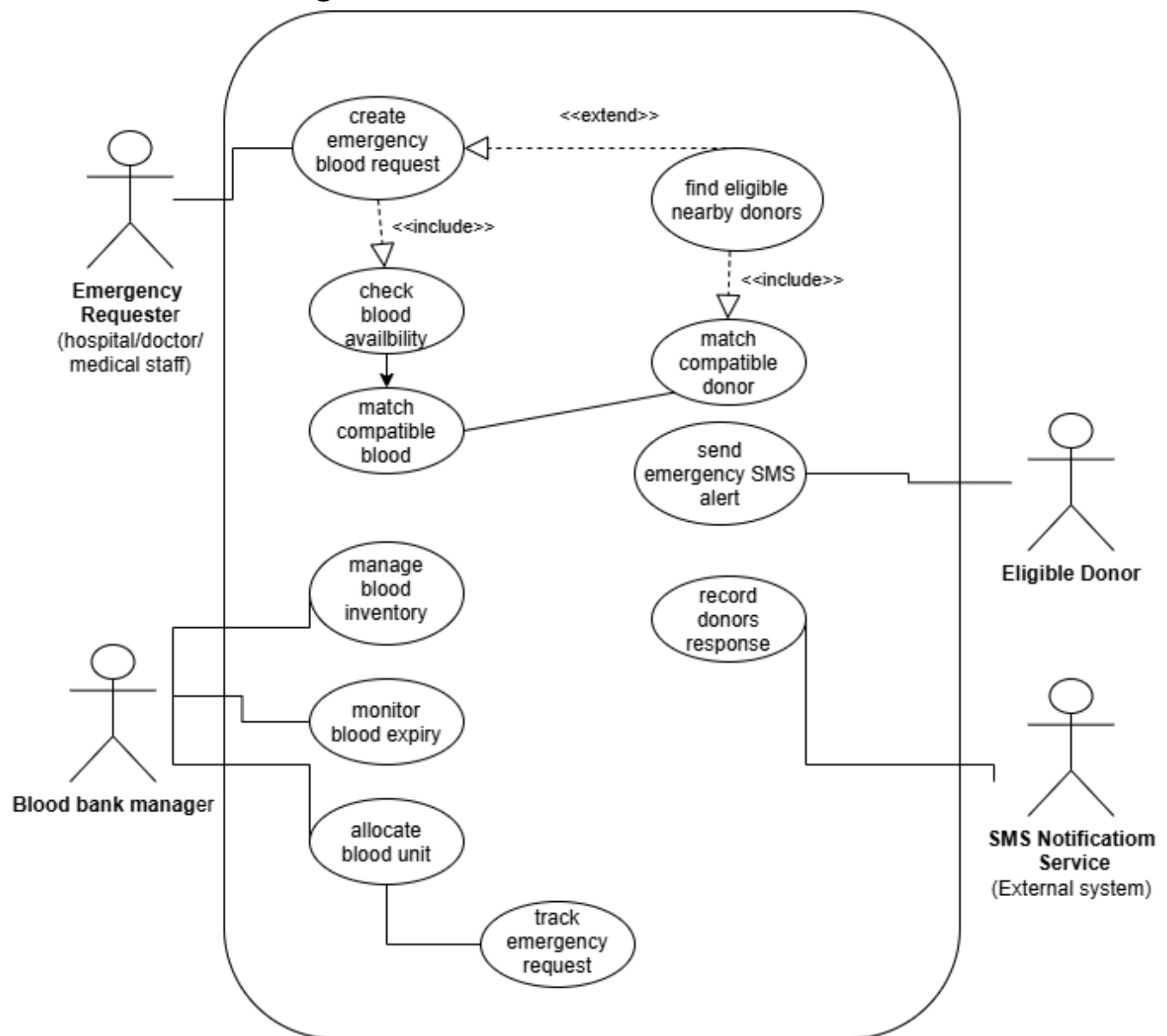
UML Relationships

«include» relationships: Create Emergency Blood Request includes Check Blood Availability; Find Eligible Nearby Donors includes Match Compatible Donor.

«extend» relationship: Send Emergency SMS Alert extends Create Emergency Blood Request when a critical shortage requires donor matching.

The UML use-case diagram should be submitted as a separate diagram image in the GitHub repository.

UML Use-Case Diagram



The diagram models the Emergency Requester, Blood Bank Manager, Eligible Donor, and SMS Notification Service, together with the primary system use cases and the required «include» and «extend» relationships.

Use-Case Flow Specification

Use Case: Handle Emergency Blood Request

Primary Actor: Emergency Requester

Supporting Actors: Blood Bank Manager, Eligible Donor, SMS Notification Service

Goal: Process an emergency blood request by checking compatible inventory and, when necessary, matching and notifying eligible nearby donors.

Preconditions

- The blood bank system is operational.
- Current blood inventory is available in the system.
- Donor information is available for matching.
- The Emergency Requester has the required emergency blood-request information.

Postconditions

- The emergency request is recorded.
- Available compatible blood units are identified and/or allocated.
- If a critical shortage exists, eligible compatible donors within the required radius are identified.
- Emergency SMS notifications are sent to matching donors.
- The request and related actions are recorded for tracking.

Main Success Scenario

- The Emergency Requester submits an emergency blood request.
- The system records the emergency request.
- The system checks the real-time blood inventory.
- The system identifies blood units compatible with the requested blood group and component.
- The system checks the availability and shelf life of matching blood units.
- If suitable blood units are available, the system reserves or allocates the required units.
- If sufficient compatible inventory is unavailable, the system initiates donor matching.
- The system identifies eligible compatible donors within a 10 km radius.
- The system sends SMS emergency alerts to the matching donors.
- The system records the notification and request status.
- The Emergency Requester receives confirmation that the emergency request has been processed.

Alternate Flow — No Eligible Donors Found

- The system searches for compatible eligible donors within the 10 km radius.
- No eligible donors are found.
- The system records that no nearby compatible donor was available.
- The Blood Bank Manager is notified of the shortage.
- The emergency request remains active for further action.